

Probability and Statistics (SH 244) Past Questions Analysis (2079 - 2082)

UNIT 1: Descriptive Statistics

Syllabus: Meaning and role of statistics in engineering, descriptive vs. inferential statistics, frequency distributions, cumulative frequency, histograms, pie diagrams, measures of location (mean, median, mode), measures of variability (range, mean deviation, standard deviation, coefficient of variation).

Short Answer Questions

2082 Semester End (Q1): Define and differentiate between descriptive and inferential statistics with example. The scores of two batsman A and B in six innings during a certain match are as follows:

Batsman A: 32, 28, 47, 63, 71, 39

Batsman B: 19, 31, 48, 53, 61, 90

Determine which of the Batsman is consistent in scoring? **[Marks: 1+4]**

2081 Semester End (Q1): Define and differentiate between descriptive and inferential statistics with particular example. An engineer tested six samples of each of three types of break for use in particular construction. The following data are compressive strength of break in (psi) unit:

Type A: 2900, 3105, 2950, 2859, 3155, 3132

Type B: 2805, 2985, 3067, 3204, 3250, 3119

Type C: 3207, 2825, 2836, 3222, 3121, 3218

- (i) Calculate the average compressive strength of each type.
- (ii) Which type of break is best and why? **[Marks: 2+6=8]**

2080 Semester End (Q1): Define and distinguish between descriptive and inferential statistics with example. The scores of two golfers for 10 rounds each are:

Golfer A: 58, 59, 60, 54, 65, 66, 52, 75, 69, 52

Golfer B: 78, 84, 56, 92, 65, 86, 78, 44, 54, 68

Which may be regarded as the more consistent player? Why? **[Marks: 2+6=8]**

2079 Semester End (Q1): Differentiate between descriptive statistics and inferential statistics and also describe their tools with examples. An analysis of monthly wages paid to workers in two firms A and B gives following results:

Metric	Firm A	Firm B
No. of workers	550	650
Average wage (Rs)	190	185
Variance of wage	81	100

(a) In which firm A or B, is there greater variability in individual wages? (b) Calculate the average monthly wages and variance of distribution of wages of all workers in both firms.

[Marks: 2+6=8]

Unit Trend Summary

Primary Focus Area: Every single paper's Question 1 starts with a theoretical definition/differentiation of Descriptive vs. Inferential Statistics (usually for 1–2 marks).

Numerical Core: The remaining marks (4–6) are always assigned to a consistency/variability analysis using the Coefficient of Variation (C.V.).

Key Pattern: You are given two raw datasets (batsmen, golfers, or compressive strengths) and must calculate:

$$\text{Mean } (\bar{x}) = (\Sigma x) / n$$

$$\text{Standard Deviation } (\sigma) = \sqrt{[\Sigma(x - \bar{x})^2 / n]}$$

$$C.V. = (\sigma / \bar{x}) \times 100\%$$

The player/item with the lower C.V. is more consistent.

Alternate Numerical: 2079 Semester End featured a combined mean and combined variance problem for two distinct groups:

$$\bar{x}_{12} = (n_1\bar{x}_1 + n_2\bar{x}_2) / (n_1 + n_2)$$

$$\sigma_{12}^2 = [n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)] / (n_1 + n_2)$$

$$\text{where } d_1 = \bar{x}_1 - \bar{x}_{12}, d_2 = \bar{x}_2 - \bar{x}_{12}$$

UNIT 2: Probability

Syllabus: Basic terminologies, sample spaces and events, counting principles, probability of an event, addition law, dependence/independence, conditional probability, multiplication law, Bayes' Theorem and its applications.

Short Answer Questions

2082 Semester End (Q2): Define conditional probability with example. If two dice are tossed together, find the probability that sum of turning faces is even, given that one face is 4. [Marks: 1+4]

2081 Semester End (Q2): With example define partition of a sample space. The probabilities that three persons A, B and C will be selected at the post of VC of FU are $\frac{3}{9}$, $\frac{4}{9}$ and $\frac{2}{9}$ respectively. The probabilities that they will introduce grade increment system if they are selected are $\frac{1}{7}$, $\frac{2}{7}$ and $\frac{3}{7}$ respectively. Find the probability that grade increment system will be introduced at university. [Marks: 2+6=8]

2081 Mid-Term (Q1): Define conditional probability with example and state Bayes' theorem. In a factory, two machines A and B produce 75% and 25% of total production respectively. Among their total output, 6% and 7% are found to be defective. Suppose that an item is selected at random and found to be defective, what is the probability that it was produced by machine A? [Marks: 8]

2080 Semester End (Q2): Define and distinguish between conditional probability and Bayes theorem. There are three candidates for the position of Principal: Mr. A, Mr. B and Mr. C, whose chances of selection are in 1:2:3 respectively. The probability that they will introduce co-education if they are selected is 0.3, 0.5, and 0.8 respectively. Find the probability that there was co-education in the college. [Marks: 2+6=8]

2079 Supplementary (Q2): Define conditional probability and Bayes theorem. In a factory, A produces 60% of total output and B produces 40% of total output. Further, 8% of output of machine A and 6% of outputs of machine B are defective. In a day's production, these two machines produce 10,000 items. If an item is selected at random and is found to be defective, what is the probability that it was produced from machine B? [Marks: 2+6=8]

2079 Semester End (Q2): Do you think that there is any difference between conditional probability and Bayes theorem? If any, describe with example. In a certain campus, 25% of boys and 10% of the girls are studying mathematics. The girls constitute 60% of students. If a student is selected at random and is studying mathematics, what is the probability that the student is a girl? [Marks: 2+6=8]

Unit Trend Summary

Primary Focus Area: This unit almost exclusively produces Bayes' Theorem or Total Probability Theorem numerical problems.

Typical Pattern: You are given 2 or 3 mutually exclusive paths (candidates Mr. A/B/C, or machines A/B, or gender split) with their prior probabilities $P(E_i)$, and the conditional probabilities of an event occurring from each path $P(A|E_i)$ (e.g., introducing co-education, being defective, or studying math).

Core Formulations:

Total Probability Theorem (to find overall probability of the event A):

$$P(A) = \sum P(E_i) \cdot P(A|E_i)$$

Bayes' Theorem (to find posterior probability of path E_k given event A has occurred):

$$P(E_k|A) = [P(E_k) \cdot P(A|E_k)] / [\sum P(E_i) \cdot P(A|E_i)]$$

Theoretical Questions: Define "conditional probability," "partition of sample space," or "Bayes' Theorem."

UNIT 3: Discrete Probability Distributions

Syllabus: Discrete random variables, probability mass function (PMF), cumulative distribution function (CDF), expected value, Binomial distribution, Poisson distribution, Negative Binomial distribution, Hypergeometric distribution.

Short Answer Questions

2082 Semester End (Q5): With example define and differentiate between Binomial, hypergeometric and negative binomial distributions. A boy is throwing stones at a target. What is the probability that his 10th throw is his 5th success, if the probability of hitting the target at any trial is 0.5? Also find $E(X)$ and $V(X)$. [Marks: 2+3]

2081 Semester End (Q3): With example define and differentiate between binomial, negative binomial and hypergeometric random variables. Suppose a coin is tossed four times and X denotes the number of heads, find the probability distribution of X. Also find the expectation of X and X^2 . [Marks: 2+4+2=8]

2081 Mid-Term (Q2): With example define and differentiate between discrete and continuous random variables and discuss the meaning of their probability distributions. Suppose a coin is tossed three times and X denotes number of heads, find the probability distribution of X. Also find mean and variance of X. [Marks: 8]

2080 Semester End (Q3): With examples describe the discrete and continuous random variable and their probability distributions. If the probability is 0.4 that steam will condense in a thin-walled aluminum tube at 10 atm pressure, find the probability that under stated condition, steam will condense in 4 of 12 such tubes. [Marks: 4+4=8]

2079 Supplementary (Q3): Define binomial experiment and binomial random variable with example. Suppose two coins are tossed together and X denotes the number of heads, find the probability distribution of X. Also calculate the probability of at most two heads. [Marks: 2+4+2=8]

2079 Semester End (Q3): Define the discrete random variable, discrete probability distribution and probability mass function with example. Two fair dice are rolled and X denotes the sum of two numbers that appear on top. Find the probability distribution of X. Also find the mean and variance of the distribution. [Marks: 2+5+1=8]

Long Answer Questions (Group B Integration)

2080 Semester End (Q11-a): With example discuss the meaning of Hypergeometric and negative binomial distribution. A boy is throwing stones at a target. What is the probability that his 10th throw is his 6th hit, if the probability of hitting the target is 0.6? [Marks: 4+4=8]

2079 Semester End (Q11-b): Define negative binomial distribution and state the condition of applicability for it. Write the similarities and differences between binomial and negative binomial distributions. A student has taken a 5-answer multiple-choice examination orally. He continues to answer questions until he gets 3 correct answers. What is the probability that he gets them in at most 7 questions if he guesses at each answer? [Marks: 2+5=7]

Unit Trend Summary

Primary Focus Area: Transitioning between Binomial and Negative Binomial problems.

Binomial Core: A simple experiment with fixed trials n and probability p :

$$P(X = x) = C(n, x) \cdot p^x \cdot (1 - p)^{n-x}$$

Negative Binomial Core: Waiting for the r th success on the x th trial (often phrased as "the 10th throw is his 5th hit"):

$$P(X = x) = C(x - 1, r - 1) \cdot p^r \cdot (1 - p)^{x-r}$$

Key Pattern: The "boy throwing stones" or "student answering questions until 3 correct answers" are repeating templates.

Expectation/Variance Calculations: Always be ready to write down complete PMF distributions for simple experiments (like throwing 3 or 4 coins, or rolling two dice and calculating the sum of faces) and calculating:

$$E(X) = \sum x \cdot P(X = x)$$

$$V(X) = E(X^2) - [E(X)]^2$$

UNIT 4: Continuous Probability Distribution

Syllabus: Continuous random variables, probability density function (PDF), expected values, normal distribution and its properties, standard normal distribution, normal approximation to binomial distribution, Gamma, Exponential, and Chi-squared distributions.

Short Answer Questions

2082 Semester End (Q3): With examples define and differentiate between discrete and continuous random variables. A continuous random variable X has the probability density function, $f(x) = c(2 - x)$ for $0 \leq x \leq 2$ and 0 elsewhere. Find the value of c and also determine the mean and variance of X . [Marks: 1+4]

2082 Semester End (Q4): With example discuss the meaning of probability distribution of a random variable X . If X denotes the number of heads when four coins are tossed together, find the probability distribution of X and also find mean and variance of X . [Marks: 1+4] (Note:

This is a discrete distribution question structurally placed under the normal distributions section in recent exam patterns).

2081 Semester End (Q4): If a random variable X has probability density function:

$$f(x) = x, \text{ for } 0 < x < 1$$

$$f(x) = 2 - x, \text{ for } 1 \leq x \leq 2$$

$$f(x) = 0, \text{ elsewhere}$$

(i) Verify that this function is a probability density function. (ii) Find $P(0.2 < x < 0.8)$. (iii) Also find the mean and variance of the density function. **[Marks: 2+2+4=8]**

2081 Semester End (Q6): With example discuss why normal approximation to binomial distribution is necessary. Suppose 20% of memory chips made in a certain plant are defective. Find the probabilities that in a lot of 100 randomly chosen chips: (i) at most 20 are defective. (ii) at least 85 are defective. (iii) between 45 and 75 are defective. **[Marks: 2+2+2+2=8]**

2081 Mid-Term (Q3): The scores of candidates in a certain test are normally distributed, with mean 500 and variance 10,000. Find the probability that the candidates receive scores: (i) between 350 and 550. (ii) above 400. (iii) below 500. **[Marks: 8]**

2081 Mid-Term (Q4): Discuss why normal approximation to binomial distribution is necessary? If a random variable has a binomial distribution with $n = 40$ and $p = 0.6$, find the probabilities that it will take the value: (i) exactly 15. (ii) less than 18. **[Marks: 8]**

2081 Mid-Term (Q6): If the probability function of a random variable X is:

$$f(x) = x, \text{ for } 0 < x < 1$$

$$f(x) = 2 - x, \text{ for } 1 \leq x \leq 2$$

$$f(x) = 0, \text{ for else}$$

(i) Verify that the function is a probability density function. (ii) Find $P(0.2 \leq X \leq 0.8)$ and also find the mean and variance of the pdf. **[Marks: 8]**

2080 Semester End (Q4): Define and differentiate between normal distribution and standard normal distribution. An electrical firm manufactures light bulbs that have a lifetime, before burn-out, that is normally distributed with mean equal to 800 hours and standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours. **[Marks: 3+5=8]**

2079 Supplementary (Q4): The daily wages of 1000 workers are normally distributed around a mean of Rs. 70 and with a standard deviation of Rs. 5. Estimate the number of workers whose daily wages will be: (i) between Rs. 70 and Rs. 72. (ii) between Rs. 69 and Rs. 72. (iii) more than Rs. 75. (iv) less than Rs. 63. **[Marks: 2+2+2+2=8]**

2079 Semester End (Q4): Define continuous random variable and its probability density function with example. The marks obtained by 300 students in mathematics are normally distributed with mean score 68 and standard deviation 3. How many students have obtained marks: (a) greater than 72. (b) less or equal to 64. (c) between 65 and 71 inclusive. **[Marks: 2+6=8]**

Long Answer Questions

2082 Semester End (Q6): Define normal distribution and standard normal distribution. Prove that the mean and variance of the standard normal distribution are 0 and 1 respectively. Also discuss why normal approximation to binomial distribution is necessary. A machine produces bolts of which 10% are defective. Find the probability that in a sample of 400 bolts produced by the machine: (i) at most 30 bolts are defective. (ii) between 30 and 50 bolts are defective. (iii) 55 or more bolts will be defective. [Marks: 1+4+1+4=10]

2081 Semester End (Q11-a): Define Normal distribution and standard Normal distribution. A random Normal variable X has mean 15 and variance 16, find: (i) $P(X \geq 12)$ (ii) $P(13 \leq X \leq 23)$ (iii) $P(0 \leq X \leq 20)$ [Marks: 2+2+2=8]

2080 Semester End (Q10-b): Define probability density function and probability distribution function. If a random variable X has function:

$$f(x) = 2e^{-2x}, \text{ for } x > 0$$

$$f(x) = 0, \text{ for } x \leq 0$$

(i) Verify that the function is a probability density. (ii) Find $P(1 < X < 3)$. (iii) Find $P(X > 0.5)$. [Marks: 2+5=7]

2079 Semester End (Q10-b): Why normal approximation to binomial distribution is necessary? Discuss your views with examples. If 20% of the memory chips made in a certain plant are defective, what is the probability that in a lot of 100 randomly chosen for inspection: (I) at most 15 will be defective. (II) between 10 and 15 inclusive. [Marks: 2+5=7]

Unit Trend Summary

Primary Focus Area: This is one of the heaviest units in the syllabus. It contains two major, highly predictable components:

Continuous PDFs (Integration): Verifying $\int_{-\infty}^{\infty} f(x) dx = 1$, finding a constant c , and integrating to find the mean $E(X) = \int x f(x) dx$ and variance.

Normal Distribution Word Problems: Standardizing the variable using $Z = (X - \mu)/\sigma$ and reading standard normal tables (e.g., bulb lifetimes, daily wages, or test scores).

Key Approximation Concept: The Normal Approximation to the Binomial is repeatedly asked (repetition in 2082, 2081, 2080, and 2079!).

Condition of applicability: Necessary when n is very large ($n \geq 30$) and neither p nor q is extremely close to 0 ($np \geq 5$ and $nq \geq 5$).

Approximation Parameters: Mean $\mu = np$, Standard Deviation $\sigma = \sqrt{npq}$.

Continuity Correction: Apply ± 0.5 adjustment. For example, $P(X \leq 30) \approx P(Y \leq 30.5)$.

UNIT 5: Joint Probability Distribution

Syllabus: Joint PMF for two discrete random variables, joint probability table, marginal PMF, joint PDF for continuous variables, marginal PDF, dependent/independent random variables, conditional probabilities, expected values, covariance, variance, Central Limit Theorem.

Short Answer Questions

2079 Supplementary (Q5): The joint probability density function for two random variables X and Y is given by:

$$f(x,y) = c(x+y)^2, \text{ for } 0 < x < 1 \text{ and } 0 < y < 1$$

$$f(x,y) = 0, \text{ for otherwise}$$

(i) Find the value of c. (ii) Find $P(0.2 < X < 0.5 \cap 0.4 < Y < 1)$. (iii) Find the marginal density function of X. **[Marks: 2+3+3=8]**

2079 Semester End (Q5): Define two-dimensional random variable and its probability mass and density function. If two random variables have joint density:

$$f(x,y) = (6/5)(x+y^2), \text{ for } 0 < x < 1, 0 < y < 1$$

$$f(x,y) = 0, \text{ for else}$$

(a) Find the marginal function of X and Y. (b) Find $P(0.2 < X < 0.5)$ and $P(0.4 < Y < 0.6)$.

[Marks: 2+6=8]

2080 Semester End (Q5): With particular example discuss the meaning of two-dimensional random variable and its probability distribution. The joint probability function of two discrete random variables X and Y is given by:

$$f(x,y) = c(2x+y), \text{ where } x, y \text{ are integers: } 0 \leq x \leq 2, 0 \leq y \leq 3$$

$$f(x,y) = 0, \text{ for otherwise}$$

(i) Find the constant c. (ii) Find $P(X \geq 1, Y \leq 2)$. (iii) Find the marginal function of X. **[Marks: 3+5=8]**

Long Answer Questions

2081 Semester End (Q11-b): Define two-dimensional random variable with example. The joint probability function of random variables X and Y is:

$$f(x,y) = 4xy, \text{ for } 0 < x < 1, 0 < y < 1$$

$$f(x,y) = 0, \text{ for otherwise}$$

(i) Verify the given function is a pdf. (ii) Find $P(x < 0.5, y < 0.8)$. **[Marks: 2+5=7]**

Unit Trend Summary

Primary Focus Area: Verifying joint functions and calculating marginal densities.

Continuous Integration Core: To find the constant c :

$$\int_0^1 \int_0^1 f(x,y) \, dx \, dy = 1$$

Marginal Densities:

$$f_x(x) = \int_{-\infty}^{\infty} f(x,y) \, dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f(x,y) \, dx$$

Key Difference: Pay close attention to whether the joint distribution is discrete (2080 Q5 uses summations over discrete integer ranges) or continuous (2079, 2081 use double integrals).

UNIT 6: Estimation

Syllabus: Meaning and importance of estimation, criteria of a good estimator, point estimation (method of moments, maximum likelihood MLE, least squares), interval estimation, confidence intervals, confidence limits for mean, proportion, and variance.

Short Answer Questions

2080 Semester End (Q6): With the help of an example related to your field, discuss the meaning and application of estimation. A sample of 700 units showed that 200 were damaged. Find 99% and 95% confidence limits for the proportion of damaged units. **[Marks: 3+5=8]**

2079 Semester End (Q6): With example define estimate, estimator and estimation. Also describe the importance of estimation in the engineering field. A researcher wants to estimate universe mean by using sampling technique. What should be the sample size when the permissible error between parameter value and sample statistic in 95% of chance will not be more than 1.5 and the population s.d. is 15. **[Marks: 2+6=8]**

Long Answer Questions

2082 Semester End (Q7): Discuss the meaning and importance of estimation in the engineering field. Also discuss the meaning of confidence interval with example. From a lot of units produced by machine A, a sample of 500 is drawn and tested for quality characteristics; it is found that 16 units are not meeting the specification. Another sample of size 100 is drawn from the lot of similar units produced by machine B and tested. In this case, only 3 units are found not meeting the specification. Obtain 95% and 99% confidence intervals for the difference of the proportion of defective units produced by the two machines. **[Marks: 1+1+4+4=10]**

2081 Semester End (Q7): Define estimation and discuss its types and also discuss the use of these statistical tools in your related field with particular example. Also mention the criteria of using Z-test and t-test for constructing the confidence interval. The breaking strength of 14 steel rods is tested and the calculated mean and s.d are 17.85 and 1.955 respectively. Find 99% and 95% confidence intervals for the average strength of all rods. **[Marks: 4+3+8=15]**

2079 Supplementary (Q7): Define the meaning of estimation and discuss with particular example how you can apply the concept of estimation in your related field. Two independent

samples of sizes 80 and 95 drawn from two normal populations have means 50 and 45 with variances 16 and 25 respectively. Obtain a 95% confidence interval for the difference of population means ($\mu_1 - \mu_2$). [Marks: 2+5+8=15]

2079 Semester End (Q10-a): Define parameter and statistics. Write the difference between point estimation and interval estimation with examples. A sample of 500 bulbs of a company showed an average life of 1400 hours with s.d. of 30 hours. Obtain 95% and 99% confidence intervals for the population mean. [Marks: 8]

Unit Trend Summary

Primary Focus Area: Differentiating point vs. interval estimation, and constructing Confidence Intervals for either:

Single Mean (μ): Using Z (for large sample $n \geq 30$) or t (for small sample $n < 30$):

$$\bar{x} \pm z_{(\alpha/2)} (s/\sqrt{n})$$

or

$$\bar{x} \pm t_{(\alpha/2, v)} (s/\sqrt{n})$$

Difference of Means ($\mu_1 - \mu_2$):

$$(\bar{x}_1 - \bar{x}_2) \pm z_{(\alpha/2)} \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

Proportions / Difference of Proportions: Refer to 2082 Q7 and 2080 Q6.

Key Formula (Sample Size for Mean estimation):

$$n = (Z_{(\alpha/2)} \cdot \sigma / E)^2$$

Featured in 2079 Semester End Q6, where $E = 1.5$, $\sigma = 15$, and $Z = 1.96$ (for 95% confidence), giving:

$$n = (1.96 \times 15 / 1.5)^2 = (19.6)^2 \approx 384.16 \rightarrow 385 \text{ samples}$$

UNIT 7: Hypothesis Testing

Syllabus: Null and alternative hypotheses, test statistics, critical regions, Type I and II errors, large sample tests (single proportion, difference of proportions, single mean, difference of means), small sample inferences (t-distribution, single mean, independent difference of means, paired difference test, F-test for comparing variances).

Short Answer Questions

2081 Semester End (Q5): Define sampling distribution of sample statistics with example. Suppose the population consists of $\{1, 3, 5, 7, 8\}$. If samples of size two are drawn without replacement, then: (i) Find the sampling distribution of the sample mean. (ii) Prove that the mean of the sampling distribution of the sample mean equals the population mean. (iii) Find the standard error of the sampling distribution of the sample mean. [Marks: 2+2+2+2=8]

2081 Mid-Term (Q5): With example define and differentiate between sample and population and also discuss the meaning of sampling distribution of sample statistic $\bar{x}$. Suppose a

population consists of $\{1,3,5\}$ and samples of size two are drawn with replacement: (i) Find the sampling distribution of the sample mean. (ii) Show that the sample mean is equal to the population mean. (iii) Find the standard error of the distribution. [Marks: 8]

2079 Supplementary (Q6): A population consists of five numbers 1, 3, 5, 7, and 9. Enumerate all possible samples of size 2 which can be drawn from the population without replacement. Show that the mean of the sample mean is equal to the population mean. [Marks: 4+4=8]

Long Answer Questions

2082 Semester End (Q8): Define and differentiate between sample and population with particular example. Also discuss the meaning of sampling distribution of sample statistic $\bar{x}$. Suppose the samples of size two and three are drawn with replacement from the populations $\{3,5,7\}$ and $\{0,3\}$ respectively. (i) Find the sampling distribution of the difference of two sample means. (ii) Prove that the mean of the sampling distribution of the difference of two sample means equals the difference of their corresponding population means. (iii) Find the standard error of the distribution. [Marks: 1+1+4+2+2=10]

2082 Semester End (Q9): Define hypothesis with example and discuss the meaning of hypothesis testing. Also discuss the importance of hypothesis testing in your related field. The life in hours of a battery is known to be normally distributed. A sample of 10 batteries has a mean life of 40.5 hours with a s.d. of 1.25 hours. Is there evidence to support the claim that battery life exceeds 40 hours at 5% risk? [Marks: 2+1+1+6=10]

2081 Semester End (Q8): Define hypothesis with example and discuss its types and differentiate between hypothesis and hypothesis testing. Machine A produces 25 defective items in a sample of 600 items and machine B produces 15 defective items in a sample of 300 items. Are these two machines significantly different in their performance? Justify the claim at 5% risk. [Marks: 5+10=15]

2080 Semester End (Q7): Define hypothesis and hypothesis testing. Is there any difference between estimation and hypothesis testing? Describe. The average weekly wages of a sample of 200 workers in industry A was Rs. 1150 with a S.D. of Rs. 100, and the weekly wages of a sample of 300 workers in industry B was Rs. 1050 with a S.D. of Rs. 50. Can we consider that the weekly wages paid by industry A are higher than those paid by B at 5%? [Marks: 2+2+1+10=15]

2080 Semester End (Q10-a): State the assumptions for using Z-test and t-test in hypothesis testing. It is claimed that both tea and coffee are equally popular. A random sample of 1200 persons contains 650 regular consumers of tea. Is the claim justified at 5% level of significance? [Marks: 3+5=8]

2079 Supplementary (Q8): Discuss the meaning of large and small sample in any statistical investigation and also discuss the meaning of hypothesis, hypothesis testing and its importance in research. A company claims that its light bulbs have a mean lifetime of 15,000 hours of continuous use with a s.d. of 120 hours. However, there has been a counter-claim by customers that a sample of 60 bulbs had a mean lifetime of 13,000 hours. Can you justify this claim at $\alpha = 0.02$? [Marks: 2+5+8=15]

2079 Supplementary (Q11): Discuss the meaning and importance of paired t-test. Blood pressures of 5 pressure patients were tested before and after prescribing a certain medicine. Test at 5% whether the medicine is effective or not:

Patient	1	2	3	4	5
BP before medicine	135	140	143	143	155
BP after medicine	120	131	128	122	150

[Marks: 3+12=15]

2079 Semester End (Q7): Do you think that estimation and hypothesis testing are essential tools of making inferences about population? Express your views with suitable examples. Two groups, each made of 9 students, were selected randomly. A course of mathematics was taught to the first group by a discovery method, and the second group of 9 students was taught the same course by an explanation method. At the end of the course, the same test was given to both groups. The first group had a mean score of 35.22 with a variance of 24.445, while the second group had a mean score of 31.56 with a variance of 20.0275. Is there a significant difference between the effectiveness of the two methods? **[Marks: 3+12=15]**

2079 Semester End (Q11-a): When do you use the paired t-test and how to apply it. A certain stimulus administered to each of 12 patients resulted in the following increases in blood pressure: 5, 2, 8, -1, 3, 0, -2, 1, 5, 0, 4, 6. Can it be concluded that the stimulus is effective in increasing blood pressure? **[Marks: 8]**

Unit Trend Summary

Primary Focus Area: This is the most critical unit in the course, consistently generating high-weight questions.

Key Categories of Tests:

Large Sample Z-Tests:

Two Means: Comparing industry wages (2080 Q7).

Two Proportions: Defective items from machines (2081 Q8).

Single Proportion: Tea/coffee popularity (2080 Q10-a).

Small Sample t-Tests:

Two Independent Means: Comparing discovery vs. explanation teaching methods (2079 Semester End Q7).

Paired t-Test (Before/After): Blood pressure medicine efficacy (2079 Supp Q11, 2079 Semester End Q11-a).

Fundamental Theory:

Standard error computations of sample distributions (especially drawing combinations of size 2 without replacement).

UNIT 8: Analysis of Variance (ANOVA)

Syllabus: *Design of experiments, meaning and assumptions of ANOVA, Completely Randomized Design (CRD), One-way ANOVA, F-distribution and F-test, Two-way ANOVA.*

Long Answer Questions

2082 Semester End (Q10): Define and differentiate between Z-test, T-test and ANOVA test. Also discuss different types of ANOVA with example. The following table represents the sales of three salesmen in three different districts:

Districts	Sale of A	Sale of B	Sale of C
Kanchanpur	14	20	16
Kailali	12	23	15
Doti	10	20	10

Test whether there is any significant difference in the sales of different districts due to salesman? [Marks: 1+2+7=10]

2081 Semester End (Q9): Define ANOVA test and discuss different types of ANOVA with example. Also discuss why ANOVA test is necessary. The following is the sales of three salesmen in different districts:

District	Sales of A	Sales of B	Sales of C
Kanchanpur	14	20	16
Kailali	12	22	14
Doti	10	18	12
Baitadi	8	16	10

Test whether there is any significant difference of sales in different districts? At 5%. [Marks: 3+12=15]

2080 Semester End (Q8): Define ANOVA test, describe its types and mention the assumptions of ANOVA test. The following are the breaking strengths of three brands of cables:

Brand	Cable breaking strengths
A	40, 30, 50, 60, 30, —
B	60, 40, 55, 65, —, —
C	60, 50, 70, 65, 75, 40

Construct ANOVA table and test for the equality of average breaking strength of cables at 5%. [Marks: 2+2+3+8=15]

2079 Supplementary (Q10): Discuss the meaning and importance of analysis of variance (ANOVA) and also discuss why it is different from Z-test and t-test. Three training methods were

compared to see if they led to greater productivity after training. The following are productivity measures for individuals trained by each method:

Method 1: 45, 40, 50, 39, 52, 44

Method 2: 59, 43, 47, 51, 39, 49

Method 3: 41, 37, 43, 40, 52, 39

Perform an analysis of variance to test at 0.05 level of significance whether the three training methods lead to different levels of productivity. **[Marks: 2+1+12=15]**

2079 Semester End (Q8): Why do we need ANOVA? Explain the weakness of Z-test and t-test. Explain briefly about one-way ANOVA with its application. The following are the breaking strengths of three different brands of cables:

Brand	Cable breaking strengths
A	40, 30, 50, 60, 30, —
B	60, 40, 55, 65, —, —
C	60, 50, 70, 65, 75, 40

Construct ANOVA table and test for the equality of average breaking strength of cables at 5%. **[Marks: 1+1+2+11=15]**

Unit Trend Summary

Primary Focus Area: Setting up the null hypothesis $H_0: \mu_1 = \mu_2 = \mu_3$ and building the ANOVA table.

Key Repeated Problem: The "breaking strength of three brands of cables (A, B, C)" is a duplicate question across 2080 and 2079 Semester End!

Core Computations (One-Way ANOVA):

Grand Total (T), Correction Factor ($CF = T^2/N$).

Total Sum of Squares ($SST = \sum X^2 - CF$).

Between-Group Sum of Squares ($SSG = \sum (T_i^2/n_i) - CF$).

Error Sum of Squares ($SSE = SST - SSG$).

Degrees of freedom ($df_{Group} = k - 1$, $df_{Error} = N - k$).

Calculate F-Statistic: $F = MSG / MSE$ and compare to critical F-value.

Two-Way ANOVA Trend: In recent papers (2082 and 2081), the table features both districts and salesmen. This requires a two-way ANOVA table (testing for row effects and column effects separately).

UNIT 9: Correlation and Regression Analysis

Syllabus: Covariance, simple/multiple/partial correlation coefficients, properties, simple regression, regression lines, scatter diagram, method of least squares, regression equations, slope/intercept confidence intervals, difference between correlation and regression.

Long Answer Questions

2082 Semester End (Q10): Define correlation and regression analysis and differentiate between them. Also discuss different types of correlation and different techniques of studying correlation between two variables with example. The amounts of chemical compound Y that dissolved in 100gm of water at various temperatures X were recorded as follows:

X (Temp)	15	30	45	60	75	10	21	33	46
Y (Chemical)	8	12	25	48	14	24	28	42	44

(i) Find the equations of regression lines. (ii) Estimate the amount of chemical that will dissolve in 100gm of water at 50°C. [Marks: 15]

2081 Semester End (Q10): (Identical to 2082 Q10 above in parameters, data, and questions). [Marks: 15]

2080 Semester End (Q9-a): Explain the concept of regression and correlation analysis with example and differentiate between them. The marks obtained by 10 students in statistics (X) and methodology (Y) are:

X	52	50	55	50	60	70	72	78	80	63
Y	20	25	60	45	80	25	55	65	25	75

Calculate the rank correlation coefficient and interpret the result. [Marks: 2+6=8]

2080 Semester End (Q11-b): The shear resistance of soil y kN m⁻² is determined by measurements as a function of the normal stress x kN m⁻² as follows:

Stress (X)	10	11	12	13	14	15	16	17	18
Resistance (Y)	14.08	15.57	16.94	17.68	18.49	19.55	20.68	21.72	22.80

Estimate the shear resistance when normal stress is 20 kN m⁻². [Marks: 7]

2079 Supplementary (Q9): Define and differentiate between correlation and regression and discuss their different types with example. Also discuss their importance in the engineering sector. The shear resistance of soil, y kN m⁻² is determined by measurement as a function of the normal stress x kN m⁻² as follows:

Stress (X)	10	11	12	13	14	15	16	17
Resistance (Y)	14.08	15.57	16.94	17.68	18.49	19.55	20.68	21.72

Estimate the approximate shear resistance of soil when normal stress is 21 kN m⁻². [Marks: 4+3+8=15]

2079 Semester End (Q9-a): Differentiate between correlation and regression with example. The marks secured by 10 students in statistics (X) and research methodology (Y) are:

Serial no.	1	2	3	4	5	6	7	8	9	10
X	52	50	55	50	60	70	72	78	80	63
Y	20	25	60	45	80	25	55	65	25	75

Calculate the rank correlation coefficient and interpret the result. **[Marks: 8]**

Unit Trend Summary

Primary Focus Area: Fitting simple linear regression equations or calculating rank correlation.

Exact Question Repetition:

The "chemical dissolving in water" regression problem was asked identically in both 2082 (Q10) and 2081 (Q10).

The "marks in statistics (X) and research methodology (Y)" rank correlation problem was asked identically in 2080 (Q9-a) and 2079 Semester End (Q9-a).

The "shear resistance of soil vs. normal stress" regression problem was asked with near-identical datasets in 2080 (Q11-b) and 2079 Supp (Q9).

Core Formulations:

Regression Line of Y on X:

$$Y - \bar{Y} = b_{yx}(X - \bar{X})$$

where:

$$b_{yx} = [n\sum XY - (\sum X)(\sum Y)] / [n\sum X^2 - (\sum X)^2]$$

Spearman's Rank Correlation Coefficient (R):

$$R = 1 - [6\sum d^2 / n(n^2 - 1)]$$

(Note: Be extremely careful when ranking identical scores, applying average rank corrections if needed).

Part 3: Study Planner & Priority Guide

Based on past mark distributions and historical frequencies, the 9 syllabus chapters are ranked from highest priority to lowest priority to optimize your revision.

Priority	Unit	Key Focus Areas	Expected Marks / Weight	Recommendation
1	Unit 7: Hypothesis Testing	Z-tests (two proportions, two means), independent t-tests, paired t-test.	20% – 25%	Highest Priority. Always leads to a 10/15-mark long question and multiple short questions.
2	Unit 4: Continuous Distributions	Normal distribution standardization word problems, continuous PDF integration, and Normal Approximation to Binomial.	15% – 20%	Highest Priority. Extremely predictable integration & standardizing math.
3	Unit 8: ANOVA	CRD, One-way and Two-way ANOVA tables, calculation of Sum of Squares, F-distribution test.	12% – 15%	Highest Priority. A single 10/15-mark question is guaranteed in every final paper.
4	Unit 9: Correlation & Regression	Spearman's rank correlation coefficient, least-squares line of regression fitting (Y on X).	10% – 15%	High Priority. High rate of duplicate questions. Practice the 'chemical solubility' and 'soil shear resistance' datasets.
5	Unit 1: Descriptive Statistics	Descriptive vs. Inferential definitions, consistency analysis using Coefficient of Variation (C.V.).	8% – 10%	Medium Priority. Standard Question 1 on every paper. Quick and easy marks.
6	Unit 2: Probability	Conditional Probability, Bayes' Theorem application (VC selection, machine defectives).	8% – 10%	Medium Priority. Practice the formula structure for conditional probability and posterior splits.
7	Unit 6: Estimation	Point vs. Interval estimation, Confidence Interval construction for single/difference of means, sample size n calculation.	8% – 10%	Medium Priority. Often integrated with Unit 7 concepts.
8	Unit 5: Joint Probability	Verifying joint densities (integrals or double sums), finding marginal functions.	5% – 8%	Low Priority. Theoretical definitions and standard double integration.
9	Unit 3: Discrete Distributions	Binomial vs. Negative Binomial distribution conditions, expected value, boy throwing stones hit probability.	5% – 8%	Low Priority. Simple PMF modeling.